

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA5112

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: Total Marks: 50

Annexure A

Code of Conduct:

I _E . Lakshmi Narayanan(192524249)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

E. Lakshmi Narayanan

Annexure A

Analytical Problem Solving

1. RSA Key Generation and Security Analysis

Given:

$p = 17, q = 19$

Compute n and $\phi(n)$

Choose $e = 5$ and calculate the private key d

Encrypt the message $M = 7$

Decrypt the ciphertext and verify correctness

Analyze the impact of choosing a small prime number on RSA security

1. RSA Key Generation and Security Analysis

Description:

RSA is a public-key cryptographic algorithm used for secure communication. It uses a public key for encryption and a private key for decryption. Its security relies on the difficulty of factoring large prime numbers.

Given: $p=17$, $q=19$, $e=5$, $M=7$

$$n = p \times q = 17 \times 19 = 323$$

$$\phi(n) = (17-1)(19-1) = 16 \times 18 = 288$$

Private key $d = 173$ (since $5 \times 173 \equiv 1 \pmod{288}$)

Public Key = $(5, 323)$

Private Key = $(173, 323)$

Encryption:

$$C = 7^5 \pmod{323} = 11$$

Decryption:

$$M = 11^{173} \pmod{323} = 7$$

Verification: Original message = Decrypted message.

Security Analysis:

Using small prime numbers makes RSA insecure because attackers can easily factor n and recover the private key. Modern RSA uses at least 2048-bit keys.

2. ElGamal Encryption Analysis

Given:

$p = 23$, $g = 5$, private key $x = 7$, random $k = 3$, message $M = 10$

Compute the public key

Perform encryption to generate ciphertext pair $(C1, C2)$

Decrypt the ciphertext to retrieve the original message

Analyze how randomness (k) improves security in ElGamal

2. ElGamal Encryption Analysis

Description:

ElGamal is an asymmetric encryption algorithm based on the Diffie–Hellman key exchange. A random value k is used during encryption, making the ciphertext different even for the same plaintext.

Given: $p=23$, $g=5$, $x=7$, $k=3$, $M=10$

Public key:

$$y = 5^7 \bmod 23 = 17$$

Public Key = (23,5,17)

Encryption:

$$C1 = 5^3 \bmod 23 = 10$$

$$C2 = 10 \times 17^3 \bmod 23 = 2$$

Ciphertext = (10,2)

Decryption:

Recovered Message = 10

Security Analysis:

Random value k ensures different ciphertexts for identical messages, improving security against attacks.

3. RSA Digital Signature Verification

Given:

$p = 11$, $q = 17$, $e = 7$, message $M = 8$

Compute n and $\phi(n)$

Find private key d

Generate the digital signature for the message

Verify the signature using the public key

Analyze how digital signatures ensure non-repudiation

3. RSA Digital Signature Verification

Description:

RSA digital signatures verify the authenticity and integrity of messages. The sender signs using the private key, and the receiver verifies using the public key.

Given: $p=11$, $q=17$, $e=7$, $M=8$

$$n = 187$$

$$\phi(n) = 160$$

Private key $d = 23$

Signature:

$$S = 8^{23} \bmod 187 = 57$$

Verification:

$$57^7 \bmod 187 = 8$$

The signature is valid.

Non-repudiation:

Only the owner of the private key can generate the signature, preventing denial of authorship.

4. Diffie-Hellman with Attack Resistance Evaluation

Given:

$$p = 29, g = 2$$

User A private key = 5

User B private key = 12

Compute public keys for both users

Compute the shared secret key

Analyze what happens if an attacker intercepts and replaces public keys

Suggest a cryptographic solution to ensure authenticity

4. Diffie-Hellman Key Exchange

Description:

The Diffie–Hellman key exchange algorithm enables two users to establish a shared secret key over an insecure communication channel. Although it provides secure key exchange, it is vulnerable to man-in-the-middle (MITM) attacks unless authentication mechanisms such as digital signatures or certificates are used.

Given:

$p=29$, $g=2$

A private key=5

B private key=12

Public Keys:

$A = 2^5 \bmod 29 = 3$

$B = 2^{12} \bmod 29 = 7$

Shared Secret:

$K = 16$

Attack Analysis:

If an attacker replaces the public keys, a Man-in-the-Middle attack becomes possible.

Solution:

Use Digital Signatures or Digital Certificates (PKI) to authenticate public keys.

5. Hybrid Encryption System Analysis

Given:

A system uses RSA for key exchange and AES for data encryption

RSA encryption time = 5 ms per key exchange

AES encryption time = 0.5 ms per data block

Total data blocks = 500

Calculate total time taken for encryption

Compare with using only RSA for encrypting all data

Analyze why hybrid encryption is more efficient

Justify its use in modern secure communication systems

5. Hybrid Encryption System Analysis

Description:

A hybrid encryption system combines the advantages of asymmetric and symmetric cryptography.

RSA is used to securely exchange the encryption key, while AES is used for encrypting large amounts of data efficiently. This combination provides both strong security and high performance, making it suitable for modern communication systems.

Given:

RSA key exchange = 5 ms

AES = 0.5 ms/block

Blocks = 500

Hybrid Time:

$$5 + (500 \times 0.5) = 255 \text{ ms}$$

RSA Only:

$$500 \times 5 = 2500 \text{ ms}$$

Comparison:

Hybrid Encryption = 255 ms

RSA Only = 2500 ms

Analysis:

Hybrid encryption is much faster because RSA encrypts only the session key, while AES encrypts the data efficiently. This approach is used in HTTPS, VPNs, banking systems, and cloud services.